

FITBUDDY

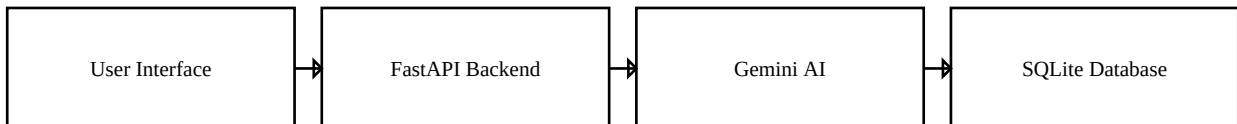
AI FITNESS PLAN GENERATOR

PHASE 3 – PROJECT DESIGN

Project Design Document

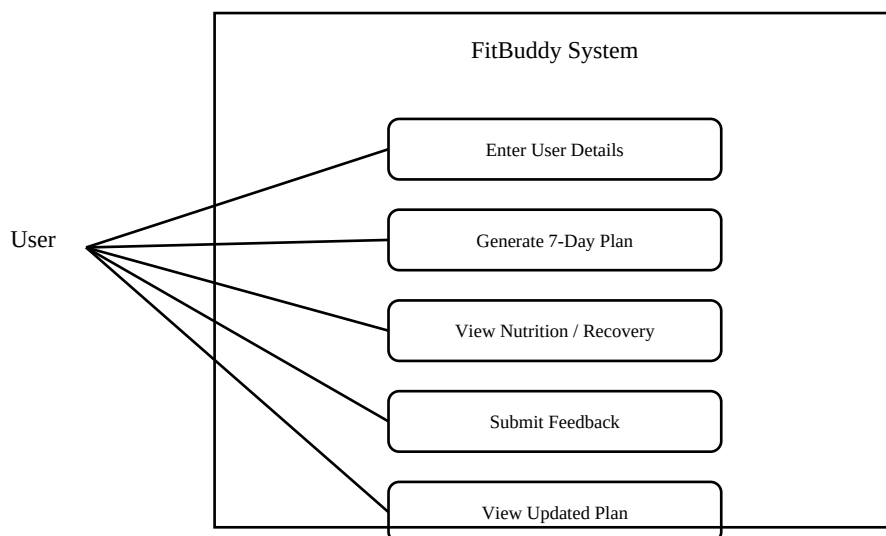
1. System Architecture Diagram

The FitBuddy system consists of a user interface, FastAPI backend, Gemini AI service and SQLite database. The backend receives user data, validates it, sends a suitable request to Gemini AI, receives generated fitness content, stores required information and displays the result.



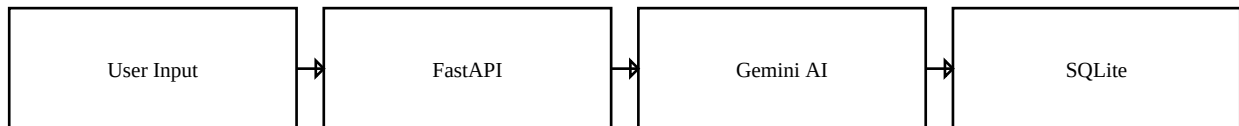
2. Use Case Diagram

The main user interactions supported by FitBuddy are entering fitness details, generating a 7-day workout plan, viewing nutrition and recovery tips, submitting feedback, receiving an updated plan, and storing/retrieving information.

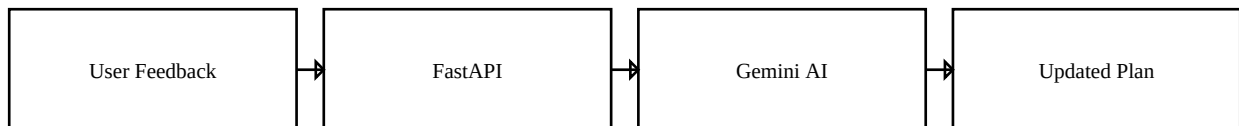


3. Data Flow Diagram (DFD)

The main data flow is: user input is sent to the FastAPI backend; the backend processes the request and uses Gemini AI to generate the fitness plan; required information is stored in SQLite; the generated plan and feedback-based updated plan are returned to the user.

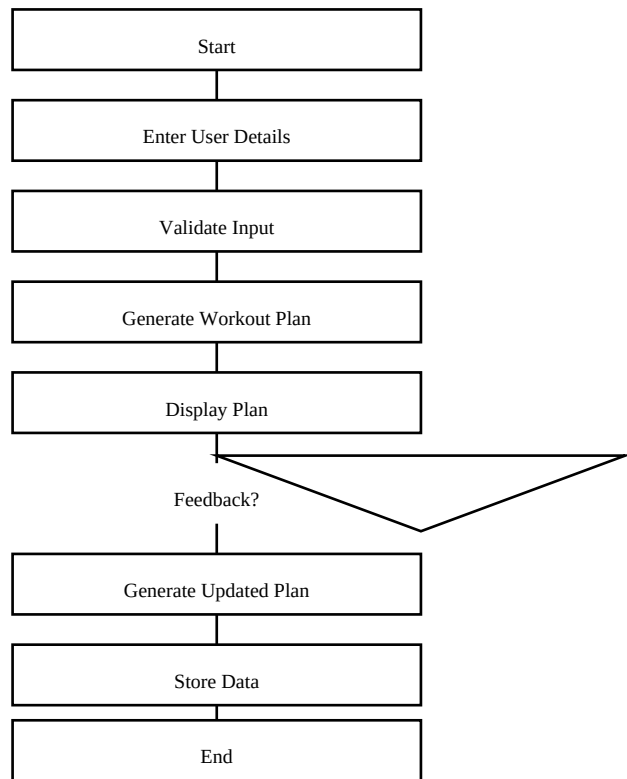


Feedback flow



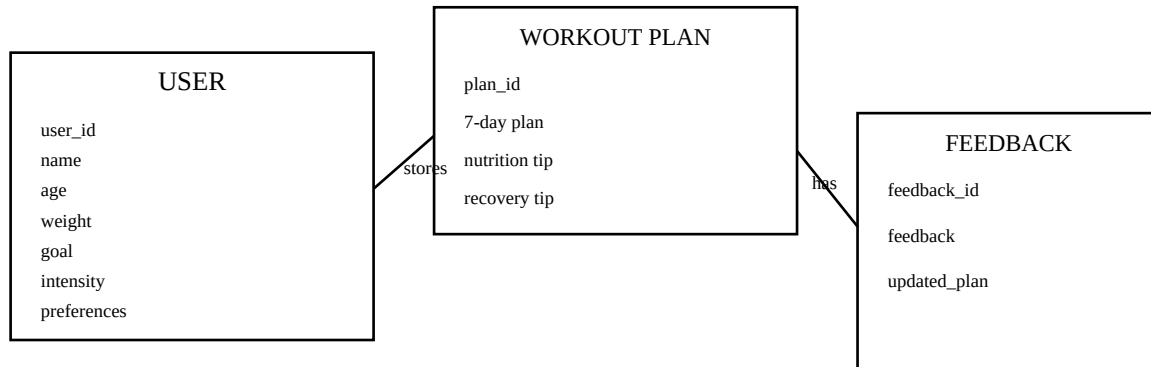
4. Flowchart

The application flow begins with user input, followed by validation and AI-based plan generation. The user can then provide feedback, after which an updated plan is generated and stored.



5. ER Diagram / Database Design

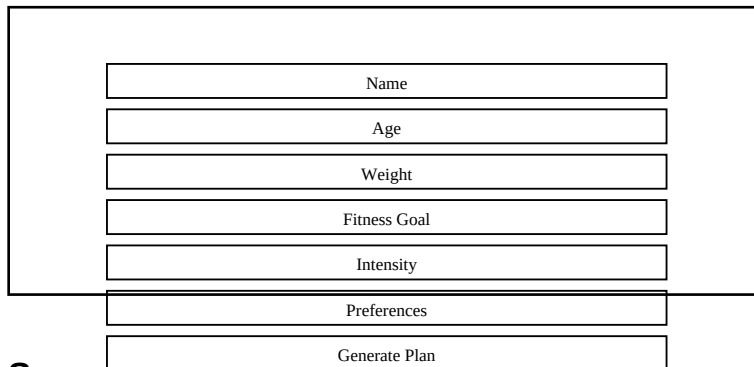
Based on the FitBuddy project description, SQLite with SQLAlchemy stores user information, generated workout plans, nutrition/recovery tips, feedback and updated plans. The design below represents the main stored user record and its plan/feedback data.



6. UI Design / Wireframes

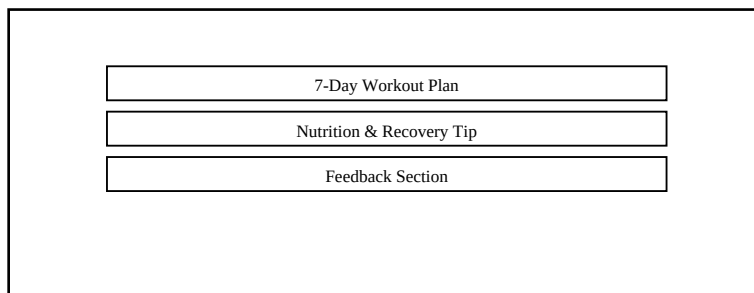
The UI design follows the application flow described in the project: user input → generated workout plan → nutrition/recovery information → feedback → updated plan.

User Input / 7-Day Fitness Plan Form



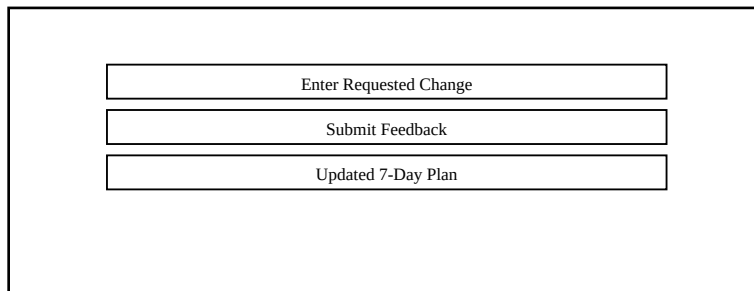
A wireframe for the 'User Input / 7-Day Fitness Plan Form'. It consists of a large rectangular container. Inside, there are seven horizontal input fields stacked vertically. The first five fields are labeled 'Name', 'Age', 'Weight', 'Fitness Goal', and 'Intensity' respectively. Below these five fields, there is a horizontal line. Below the line, there are two more horizontal input fields labeled 'Preferences' and 'Generate Plan'.

Generated Plan Screen



A wireframe for the 'Generated Plan Screen'. It consists of a large rectangular container. Inside, there are three horizontal input fields stacked vertically. The first field is labeled '7-Day Workout Plan', the second is labeled 'Nutrition & Recovery Tip', and the third is labeled 'Feedback Section'.

Feedback / Updated Plan Screen



A wireframe for the 'Feedback / Updated Plan Screen'. It consists of a large rectangular container. Inside, there are three horizontal input fields stacked vertically. The first field is labeled 'Enter Requested Change', the second is labeled 'Submit Feedback', and the third is labeled 'Updated 7-Day Plan'.

7. Design Summary

The Phase 3 design organizes the FitBuddy application into its major architecture, user interactions, data flow, application flow, database design and interface layout. These design elements provide the structure used for the subsequent development and testing phases.